21	Two metal wires X and Y are connected in series and a current is flowing through them. The number of free electrons per cubic metre in wire X is twice that in wire Y. The diameter of the wire X is half that of wire Y. The drift speed of the free electrons in wire X is 0.52 mm s^{-1}. What is the drift speed of the free electrons in wire Y?							
	A	0.26 mm s^{-1}	**B**	0.52 mm s^{-1}	**C**	1.04 mm s^{-1}	**D**	2.08 mm s^{-1}
L2	**Answer: A** **Current in wire X** $I_X = e\, n_X A_X v_X = e\, n_X (\pi d_X^2/4)\, v_X$							

[Turn over

Current in wire Y

$I_Y = e\, n_Y A_Y v_Y = e\, n_Y (\pi d_Y^2/4)\, v_Y$

In series $\Longrightarrow$ Same current in X and Y

$e\, n_X (\pi d_X^2/4)\, v_x = e\, n_Y (\pi d_Y^2/4)\, v_Y$

$n_X d_X^2 v_x = n_Y d_Y^2 v_Y$

$$v_Y = \frac{n_X d_X{}^2 v_X}{n_Y d_Y{}^2} = \left(\frac{n_X}{n_Y}\right)\left(\frac{d_X{}^2}{d_Y{}^2}\right) v_X$$

$$= \left(\frac{2}{1}\right)\left(\frac{1}{2}\right)^2 \ 0.52 \text{ mm s}^{-1}$$

$$= 0.26 \text{ mm s}^{-1}$$